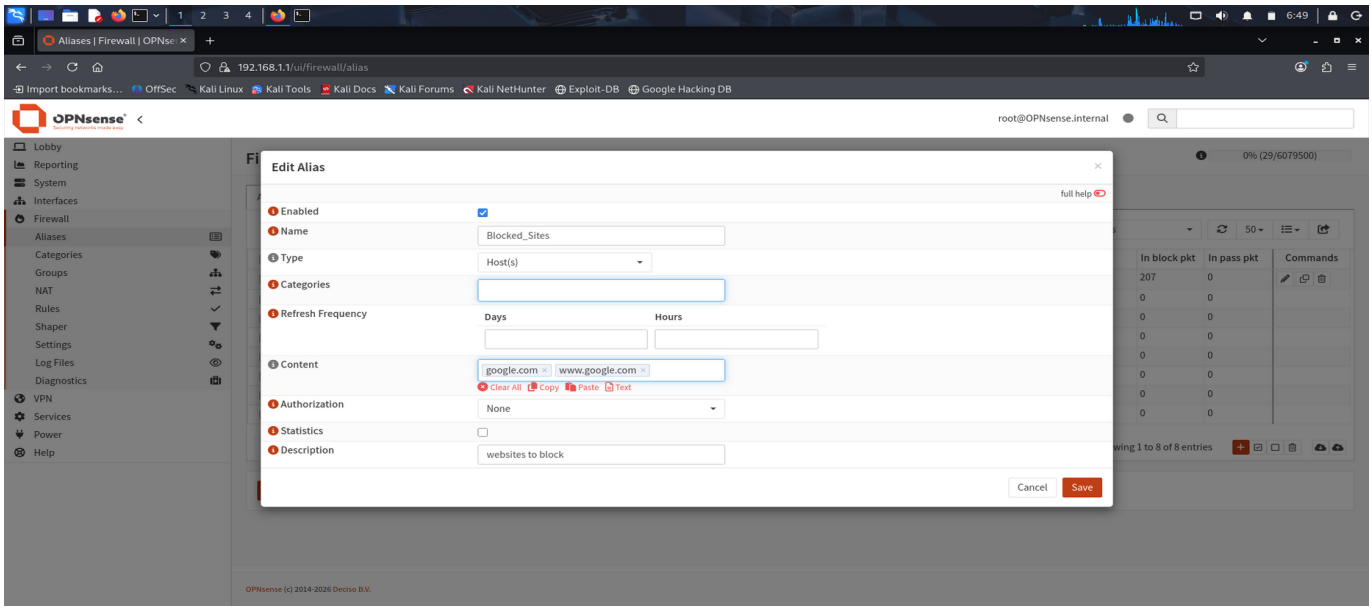
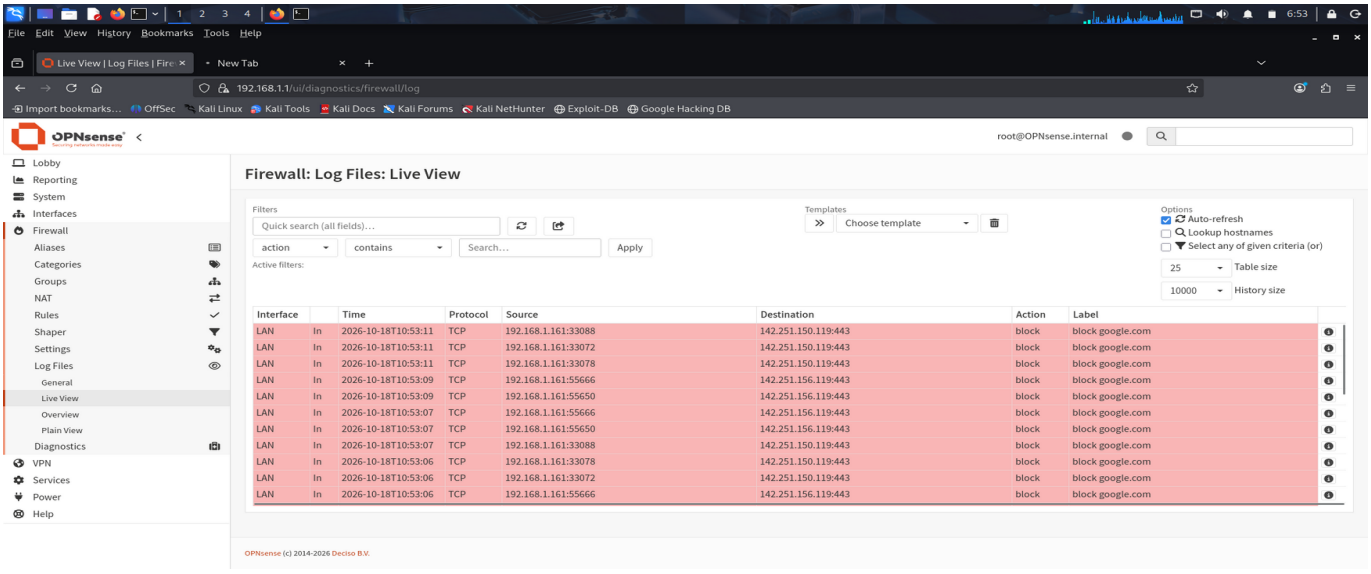
FIREWALL AND NETWORK ADDRESS TRANSLATION ASSIGNMENT

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

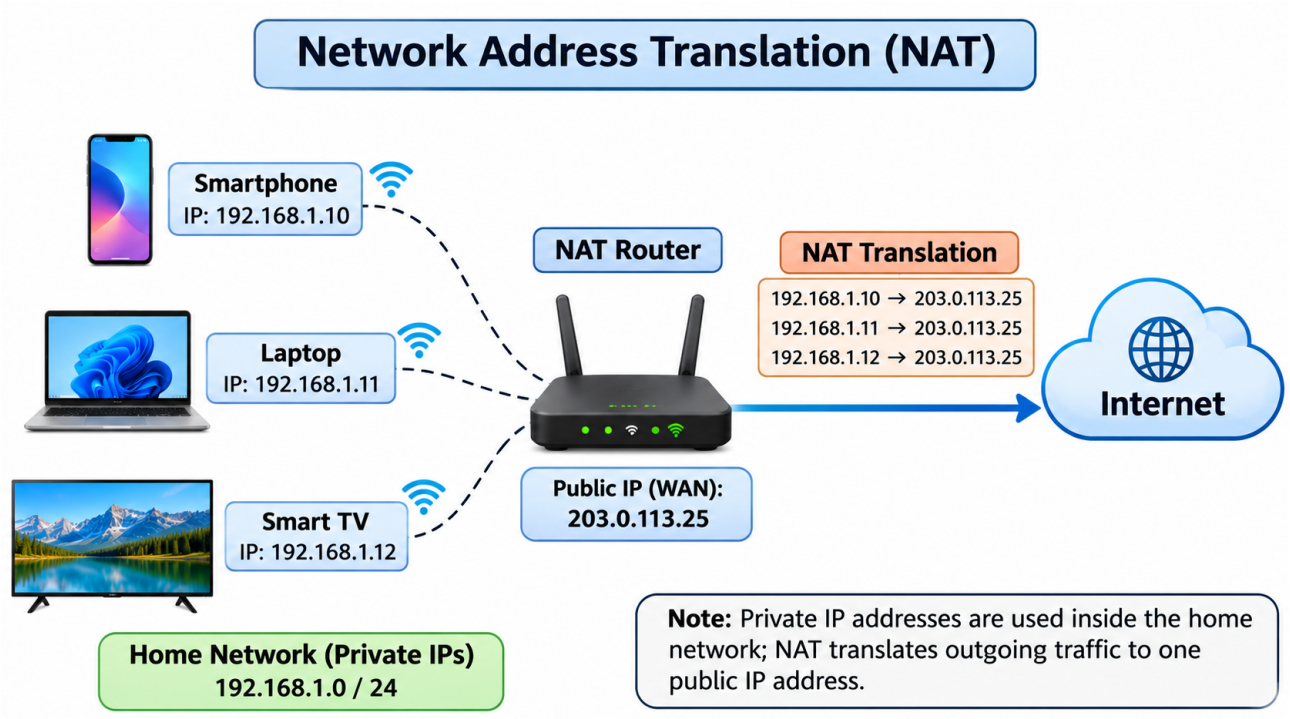
Steps to create the inbound firewall rule

Step	Instruction
1	Open Start, search for Windows Defender Firewall with Advanced Security, and open it.
2	Select Inbound Rules, then click New Rule.
3	Choose Port as the rule type.
4	Select TCP, choose Specific local ports, enter 8080, and click Next.
5	Select Block the connection.
6	Keep the Domain, Private and Public profiles selected as required.
7	Name the rule Block Incoming Port 8080 and click Finish.





2. Simulate NAT (Network Address Translation) by showing how a home Wi-Fi router assigns private IP addresses to a phone, laptop and smart TV, and translates them to a single public IP for Internet access.



3. Configure port filtering using your Wi-Fi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

General steps to configure outgoing port filtering

Step	Instruction
1	Connect a laptop to the router and open a browser.
2	Enter the router's gateway address (commonly 192.168.0.1 or 192.168.1.1) and sign in to the administrator panel.
3	Look for Security, Firewall, Access Control or IP/Port Filtering settings. Menu names vary by router.
4	Create an outgoing rule and select TCP as the protocol.
5	Set the destination port to 25, choose Deny or Block, and apply the rule.
6	Save the settings and test that outgoing SMTP traffic on port 25 is blocked.

4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Packet	Source IP	Destination IP:Port	Service	Decision
1	192.168.1.5	172.217.166.78:80	HTTP	Allow
2	10.0.0.10	34.201.23.1:22	SSH	Block
3	192.168.1.7	8.8.8.8:53	DNS	Allow

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Method	How it could protect the platform
1. Block suspicious IP addresses and traffic	A network firewall can filter packets from known malicious IP addresses, invalid sources or unexpected ports before they reach public-facing servers. This can reduce unwanted connection attempts and some forms of network scanning.
2. Protect against traffic floods and automated abuse	Firewalls and web application protection services can apply traffic thresholds and rate limits to reduce the impact of denial-of-service attempts, repeated login attempts, scraping and abusive automated requests.
3. Separate public servers from internal systems	Network segmentation and firewall rules can restrict communication between public web servers, application services, databases and administrative systems. If one server is compromised, these restrictions can limit an attacker's movement to other parts of the infrastructure.

References

OWASP – Denial of Service Cheat Sheet:
https://cheatsheetseries.owasp.org/cheatsheets/Denial_of_Service_Cheat_Sheet.html

OWASP – Network Segmentation Cheat Sheet:
https://cheatsheetseries.owasp.org/cheatsheets/Network_Segmentation_Cheat_Sheet.html

OWASP – Bot Management and Anti-Automation:
https://cheatsheetseries.owasp.org/cheatsheets/Bot_Management_and_Anti-Automation_Cheat_Sheet.html

Zomato – Draft Red Herring Prospectus (security and risk disclosures):
https://d2un9pqbzgw43g.cloudfront.net/main/Zomato-DRHP-April-27-2021-1.pdf